

Mayungbo Melvyn Oluwatobi

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GitHub: github.com/MelvTheGoat | Lagos, Nigeria

I am a Machine Learning & AI Engineer with a Statistics background, specializing in systems where the probability must be trustworthy and not merely the label. Experience spans classical ML and deep learning — forecasting, credit risk, and fraud pipelines through to sequence models and deployed retrieval-augmented generation systems — with consistent depth in temporally honest validation, calibrated decisioning, evaluation design, containerized serving, and drift monitoring.

EDUCATION

University of Ibadan

January 2021 – March 2025

B.Sc. in Statistics — Oyo, Nigeria

SQI College of ICT

July 2025 – Present

Professional Diploma in Artificial Intelligence — Ibadan, Nigeria

TECHNICAL PROJECTS

Nigerian Fintech Compliance RAG Assistant

GitHub

Tech Stack: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Built and deployed a regulatory question-answering system over Nigerian CBN circulars and the NDPA, combining sparse (BM25) and dense (ONNX-served MiniLM) retrieval fused via reciprocal rank fusion, achieving 0.96 recall@5 and 0.842 MRR against a 50-question labelled golden set with zero retrieval misses in the top 10.
- Engineered section-aware chunking that never merges across a document's own structural boundaries, keeping every citation attributable to exactly one clause, and validated the approach against a structure-blind fixed-window baseline rather than asserting it was better.
- Designed generation to fail safely under uncertainty: a sentinel-token refusal mechanism enforced across the prompt, output guardrail, and evaluation harness, plus post-generation citation verification that strips any marker the model cited but never actually retrieved.
- Ran retrieval and generation evaluation as fully separate metrics to localize failure by component; diagnosed a 0.40 refusal-correctness result to a lexical-overlap limitation in the offline stub judge specifically, rather than reporting it as deployed-system behavior — flagging re-validation against a live provider as the explicit next step instead of overstating readiness.
- Deployed on GCP Cloud Run via Docker within a ~225MB peak memory footprint (2GB budget), with per-session and daily request caps enforced in application code to bound API cost on a public endpoint.

Self-Operating Demand Forecasting Platform

GitHub

Tech Stack: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.
- Modelled asymmetric business cost by penalising unmet demand at 3× idle supply and shipping the cost-optimal quantile rather than the median, cutting expected cost by 25% on an identical model and beating a seasonal-naive baseline by 32%.
- Validated with rolling-origin backtesting across 6 expanding-window folds against seasonal-naive and historical-mean baselines, reporting MASE and pinball loss, with automated leakage tests that fail the build if any feature reads past the forecast origin.
- Implemented drift-triggered retraining using PSI and KS drift plus rolling MASE thresholds, with champion/challenger promotion gated on a 2% out-of-sample cost margin to keep degraded models out of production.

Sequence-Based Transaction Fraud Detection System

GitHub

Tech Stack: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Built and compared three deep sequence architectures (GRU, Temporal Convolutional Network, and a causally-masked Transformer encoder) over customer transaction histories, using learned embeddings for high-cardinality merchant (~2.9k vocabulary), MCC, and device-type categoricals; all models kept under 115k parameters and CPU-trainable in under 6 minutes.
- Benchmarked all architectures against a strong LightGBM baseline on engineered velocity and deviation features, reporting PR-AUC and precision at fixed review capacity (top 0.1%, 0.5%, 1%) across 3 seeds with a per-attack-type breakdown covering card testing, account takeover, and merchant compromise; sequence models cut expected cost per transaction 27% (0.0665 vs 0.0906) under an explicit cost model, though the margin came entirely from robustness to injected adversarial drift — pre-drift the baseline was marginally ahead (0.956 vs 0.950 PR-AUC).
- Addressed extreme class imbalance via focal loss and weighted sampling, isolating their distinct failure modes: focal loss degraded probability calibration (ECE 0.0063) while weighted sampling instead cost ranking quality (-0.042 PR-AUC). Corrected calibration with temperature scaling and isotonic regression, reducing ECE to 0.0005.
- Deployed real-time scoring by exporting to ONNX Runtime (2.9× faster than eager PyTorch at batch size 1) and serving via FastAPI, achieving 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag

Credit Risk Decisioning & Fairness Audit System

GitHub

Tech Stack: Python, LightGBM, scikit-learn, SHAP, WOE Scorecards, FastAPI, Docker

- Developed a full credit decisioning system producing calibrated default probabilities, cost-optimal approve/decline cutoffs, adverse action reason codes, and an append-only audit trail enabling any individual decision to be reconstructed months later.
- Benchmarked gradient boosting against a traditional WOE-binned logistic scorecard, prioritizing calibration quality (Brier score, reliability diagrams, expected calibration error) over ranking metrics because expected loss requires true probabilities rather than rankings.
- Implemented reject inference to correct the selection bias arising from outcomes being observed only for previously approved applicants, validating each correction against simulated data where true rejected-applicant outcomes were known.
- Audited fairness across sex, age, education, and marital status using demographic parity and equal opportunity differences plus within-group calibration, and presented the accuracy-fairness tradeoff curve as a policy decision rather than silently optimizing it away.

Uplift Modeling & Causal Targeting Study

GitHub

Tech Stack: Python, EconML/CausalML, LightGBM, scikit-learn, Experiment Design

- Estimated heterogeneous treatment effects across a 64,000-record randomized marketing trial to identify customers who convert because of an intervention rather than customers who convert regardless, which conventional propensity models cannot distinguish.
- Implemented S-, T-, and X-learners alongside a causal forest, validating every estimator against simulated data with known individual treatment effects, since true counterfactuals are unobservable and no accuracy metric exists on real data.
- Evaluated using Qini curves, AUUC, and decile uplift tables rather than AUC, and applied placebo, covariate-balance, and seed-stability tests; the placebo test showed the ranking was indistinguishable from noise on the primary arm, and the 18.6% of customers flagged as negative-uplift measured +0.44pp against the randomised holdout — a negative result reported rather than shipped.
- Delivered a two-page decision memo containing a budget-constrained targeting policy, a power and minimum-detectable-effect calculation for the confirmatory experiment, and an explicit statement of the assumptions that would invalidate the recommendation.

EXPERIENCE

SQI College of ICT

March 2026 – Present

Machine Learning Instructor — Ibadan, Nigeria

- Teaching machine learning fundamentals to monthly cohorts of students, focusing on predictive modeling, algorithm selection, and model evaluation metrics.
- Instructing students on data analysis, feature engineering, and model building leveraging libraries such as Scikit-Learn, PyTorch, Pandas, and NumPy.
- Guiding learners through end-to-end ML projects on real-world datasets, translating technical

Thrive Protocol

January 2025 – May 2025

Location Data Guardian & Reviewer — Remote

- Reviewed and validated large-scale location datasets for accuracy, consistency, and formatting.
- Maintained high-quality, normalized datasets optimized for downstream algorithmic analysis.

Nigerian Institute of Social and Economic Research (NISER)

May 2024 – July 2024

Internal Auditor — Ibadan, Nigeria

- Analyzed financial and operational records to identify inconsistencies and irregularities, performing anomaly detection manually in a context where false positives carried reputational cost.

Wakandanz Gaming

September 2020 – December 2021

Manager — Lagos, Nigeria

- Managed team operations, applying performance data to track outcomes and improve decision-making.

TECHNICAL SKILLS

Languages & Core Tools: Python, SQL, Bash, Git, GitHub, Docker, FastAPI, Streamlit, GitHub Actions, CI/CD, Pytest, Ruff, Mypy

AI & Machine Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, Deep Learning, Neural Networks, Sequence Models (GRU/LSTM, TCN, Transformers), Predictive Modeling, Model Calibration, LLMs, Retrieval-Augmented Generation (RAG), Hybrid Retrieval (BM25, Dense Embeddings, Reciprocal Rank Fusion), LLM Evaluation & Guardrails, Hugging Face, SHAP

Statistics & Causal Inference: Statistical Inference, Experiment Design, Power Analysis, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Time-Series Backtesting, Logistic and Poisson Regression, WOE Scorecards

Data Engineering & MLOps: Pandas, NumPy, Prefect, dbt-core, DuckDB, MLflow, Evidently, ONNX Runtime, Data Pipelines, Feature Engineering, API Development, Model Serving, Data Drift Monitoring, Vector Databases, Google Cloud Platform (Cloud Run)

AWARDS & CERTIFICATES

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action